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Brief paper

A distributed algorithm for solving mixed equilibrium problems ☆

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ABSTRACT

In this paper, the mixed equilibrium problem is solved by a multi-agent network. The objective for agents is to cooperatively find a point in a convex set, at which the sum of some local bifunctions with a free variable is non-negative. To address this problem, we propose a distributed extragradient algorithm based on a consensus strategy. By implementing the algorithm, each agent adjusts its state value by only using its own bifunction information and the local state information received from its immediate neighbors. Under mild conditions on the graph and bifunctions, it is shown that all agents reach agreement asymptotically, and the consensus state is a solution to the equilibrium problem. A simulation example is presented to demonstrate the effectiveness of our theoretical results.

1. Introduction

In equilibrium problems (EPs), the objective is to find a point at which a bifunction with a free variable is non-negative (Su, 2005). EPs have wide applications in electrical networks, market behaviors and economics (Addi, Brogliato, & Goeleven, 2011; Goeleven, 2008; Kumam & Jaiboon, 2009; Kumam, Jaiboon, Kumam, & Singta, 2010; Santos & Scheimberg, 2011; Su, 2005). For example, in the analysis of ampere–volt characteristics for circuits (formed by the interconnection of electrical devices like generators, resistors, inductors and diodes), based on Kirchhoff's laws, currents in m loops can be computed by finding $\mathbf{x} \in \Omega$ such that (refer to Addi et al. (2011) and Goeleven (2008) for details)

$$\langle \mathbf{Px} + \mathbf{U}, \mathbf{y} - \mathbf{x} \rangle + \Phi(\mathbf{y}) - \Phi(\mathbf{x}) \geq 0, \ \forall \mathbf{y} \in \Omega \tag{1}$$

where $\Omega \subseteq \mathbb{R}^m$, $\Phi : \mathbb{R}^m \to \mathbb{R}$ is the electrical potential of diodes, Φ is usually a nonlinear function, $\mathbf{P} \in \mathbb{R}^{m \times m}$ and $\mathbf{U} \in \mathbb{R}^m$ are determined by the resistance and voltage source in the circuit, respectively. Solving (1) is a typical EP, which is called a "variational inequality of the second kind" in Addi et al. (2011) and Goeleven (2008). In some large-scale electrical systems (Paatero & Lund, 2007), the electrical potential, resistance and voltage source are determined by a large number of electrical devices. In such scenarios, the bifunction is formed by the sum of multiple bifunctions. Accordingly, such an EP is referred to as a mixed equilibrium problem (MEP) (Kumam & Jaiboon, 2009; Kumam et al., 2010). In Kumam and Jaiboon (2009) and Kumam et al. (2010), MEPs with two bifunctions have been studied by centralized approaches. However, in many applications, especially in large-scale electrical networks and sensor networks (Olfati-Saber & Shamma, 2015; Paatero & Lund, 2007), decision makers can only have access to information of themselves, as well as information received from their neighbors. In such cases, a distributed strategy is often desired. Hence, we consider distributed algorithms for EPs in this paper. Moreover, taking the advantage of multi-agent networks (Lu, Jing, & Wang, 2019; Ren & Beard, 2005; Wang, Shi, Chu, Zhang, & Zhang, 2004), we will consider a more general MEP where the bifunction is formed by an arbitrarily finite number of local bifunctions.

In recent years, distributed optimization has been widely investigated via multi-agent networks (Di Lorenzo & Scutari, 2016; Jakovetić, Xavier, & Moura, 2014; Lu, Jing, & Wang, 2018; Mou, Liu, & Morse, 2015; Nedić & Olshevsky, 2015; Nedić, Ozdaglar, & Parrilo, 2010; Xi, Xin, & Khan, 2018; Zhu & Martnez, 2012). In distributed optimization, the global cost function is formed by multiple local cost functions, where all local cost functions are coupled through a common vector (Jakovetić et al., 2014; Nedić et al., 2010). To find the global minimizer of the global cost function, each agent minimizes its own cost function while exchanging local information with other agents via a communication graph.

Compared with optimization problems, EPs are more general. Moreover, EPs also cover variational inequality problems (Su, 2005). For example, the inequality in (1) is actually a combination

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of a variational inequality problem and an optimization problem. More precisely, if $\Phi(\mathbf{x})$ is linear with respect to $\mathbf{x}$, then, (1) is reduced to be a variational inequality problem. If $\mathbf{P} = 0$, then, it is an optimization problem. Note that if $\Phi(\mathbf{x})$ is nonlinear with respect to $\mathbf{x}$ and $\mathbf{P} \neq 0$, then, inequality (1) is neither an optimization problem nor a variational inequality problem. Inspired by works on distributed optimization, we wish to study MEPs in a distributed manner. Different from the investigations (Di Lorenzo & Scutari, 2016; Jakovetić et al., 2014; Lu et al., 2018; Mou et al., 2015; Nedić & Olshevsky, 2015; Nedić et al., 2010; Xi et al., 2018; Zhu & Martnez, 2012) associated with distributed optimization where the cost functions are single valued, in MEPs, one should deal with bifunctions coupled through a pair of variables, where the two variables may be inseparable. One of the variables is freely changeable and the bifunctions may even be nonlinear. This brings challenges in solving MEPs.

In this paper, we will propose a distributed extragradient algorithm for a group of agents to cooperatively solve a class of MEPs. By implementing the algorithm, each agent adjusts its state value by solving two auxiliary strongly convex optimization problems involving its own bifunction information and the local state information received from its immediate neighbors. The underlying communication graph is modeled as a time-varying directed graph. We show that under the presented algorithm, if the time-varying directed graph is balanced and uniformly strongly connected, then the multi-agent system (MAS) asymptotically reaches consensus, and the consensus state is a solution to the MEP. The result is proved based on graph theory, convex analysis theory and Lyapunov stability theory. Particularly, in dealing with the free variable in bifunctions, a Lipschitz continuity-like condition and a monotonicity-like condition are adopted. A simulation example is provided to demonstrate the effectiveness of our theoretical results.

The remainder of this paper is organized as follows. In Section 2, we formulate the problem to be studied and present the distributed algorithm for solving the MEP. In Section 3, we state our main results and give theoretical proofs in detail. In Section 4, a numerical simulation is presented. Section 5 concludes the paper.

Notations. Throughout this paper, $\mathbb{R}$ and $\mathbb{N}$ denote the set of real numbers and set of non-negative integers, respectively. Let $\mathbb{R}^m$ be the m-dimensional real vector space. For a given vector $\mathbf{x} \in \mathbb{R}^m$, $\|\mathbf{x}\|$ denotes the standard Euclidean norm of x, i.e., $\|x\| = \sqrt{x^T x}$. For function $f(\cdot) : \mathbb{R}^m \to \mathbb{R}$, we use $\partial f(\mathbf{x})$ to represent its (sub)gradient at $\mathbf{x} \in \mathbb{R}^m$. For bifunction $g(\cdot,\cdot) : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}$, we use $\partial_1 g(\mathbf{x}, \mathbf{y})$ to denote its subgradient with respect to the first argument at $\mathbf{x} \in \mathbb{R}^m$, and use $\partial_2 g(\mathbf{x}, \mathbf{y})$ to denote its subgradient with respect to the second argument at $\mathbf{y} \in \mathbb{R}^m$.

2. Preliminaries

2.1. Graph theory

The time-varying directed communication topology is denoted by $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), \mathbf{A}(t))$. $\mathcal{V}$ is a set of vertices, denoting the group of agents. $\mathcal{E}(t) \subset \mathcal{V} \times \mathcal{V}$ is an edge set, the existence of each edge between two agents implies that they can communication with each other. The weighted matrix $\mathbf{A}(t) = (a_{ij}(t))_{n\times n}$ is a non-negative matrix for adjacency weights of edges such that $a_{ij}(t) > \eta$ for some $\eta > 0$ if $(j, i) \in \mathcal{E}(t)$ and $a_{ij}(t) = 0$ otherwise. Denote $\mathcal{N}_i(t) = \{j \in \mathcal{V} | (j, i) \in \mathcal{E}(t)\}$ as the neighbor set of agent i at time t. Here we assume $i \in \mathcal{N}_i(t)$ for all $i \in \mathcal{V}$ and $t \geq 0$. For a fixed topology $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, a path of length r from node i_1 to node i_{r+1} is a sequence of $r + 1$ distinct nodes $i_1 \cdots, i_{r+1}$ such that $(i_q, i_{q+1}) \in \mathcal{E}$ for $q = 1, \ldots, r$. If there exists a path between any two nodes, then $\mathcal{G}$ is said to be strongly connected. For time-varying graph $\mathcal{G}(t)$, an B-edge set is defined as $\mathcal{E}_B(t) = \cup_{k=tB}^{(t+1)B-1} \mathcal{E}(k)$ for some constant $B > 0$. $\mathcal{G}(t)$ is $B-$strongly connected (uniformly strongly connected) if the directed graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}_B(t)$ is strongly connected for any $t \geq 0$.

Here we make the following assumption for the communication graph, which has also been used in Di Lorenzo and Scutari (2016), Nedić et al. (2010) and Zhu and Martnez (2012).

Assumption 1. $\mathbf{A}(t)$ is a doubly stochastic matrix for any $t \geq 0$, which implies that $\mathcal{G}(t)$ is balanced. Moreover, $\mathcal{G}(t)$ is $B-$strongly connected.

In the literature, several choices to satisfy double stochasticity condition have been proposed, such as the uniform weights (Blondel, Hendrickx, Olshevsky, & Tsitsiklis, 2005) and the least-mean square consensus weight rules (Xiao, Boyd, & Kim, 2007). In Gharesifard and Cortés (2010), it shows that there exists a class of weights such that the weight matrix of a strongly connected directed graph with enough number of self-loops is doubly stochastic (refer to Corollary 4.2 in Gharesifard and Cortés (2010)). Furthermore, in Gharesifard and Cortés (2012), some distributed strategies were designed for strongly connected directed graphs to accomplish a balanced/doubly stochastic assignment in finite time.

2.2. Mixed equilibrium problems

Definition 1 (*Kumam & Jaiboon, 2009; Su, 2005*)**.** Let $\Omega \subset \mathbb{R}^m$ be a closed convex set and $g(\cdot,\cdot) : \Omega \times \Omega \to \mathbb{R}$ be a bifunction, the problem of finding a point $\mathbf{x}^* \in \Omega$ such that $g(\mathbf{x}^*, \mathbf{y}) \geq 0,\ \forall\, y \in \Omega$ is called an EP. If $g(\cdot,\cdot)$ is formed by multiple bifunctions, then it is referred to as an MEP.

The objective of this paper is to distributively find a feasible solution to the following MEP:

$$\begin{aligned} &\text{Find}\ \ \mathbf{x}^* \in \Omega \\ &s.t.\ g(\mathbf{x}^*, \mathbf{y}) \geq 0, \forall\, \mathbf{y} \in \Omega \end{aligned} \tag{2}$$

where $g(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n g_i(\mathbf{x}^*, \mathbf{y})$, $\Omega \subset \mathbb{R}^m$ is a closed convex set, and each bifunction $g_i(\cdot,\cdot) : \Omega \times \Omega \to \mathbb{R}$ is convex with respect to the second argument, and g_i is only available to agent i. The convexity implies that for any $\mathbf{x}, \mathbf{y} \in \Omega$, the subgradient of function $g_i(\mathbf{x}, \mathbf{y})$ with respect to the second argument, denoted by $\partial_2 g_i(\mathbf{x}, \mathbf{y})$, always exists, and the following inequality holds:

$$g_i(\mathbf{x}, \mathbf{y}) - g_i(\mathbf{x}, \mathbf{z}) \geq \langle \partial_2 g_i(\mathbf{x}, \mathbf{z}), \mathbf{y} - \mathbf{z}\rangle,\ \ \forall\, \mathbf{x}, \mathbf{y}, \mathbf{z} \in \Omega.$$

Remark 1. Let $g_1(\mathbf{x}, \mathbf{y}) = \theta(\mathbf{x}, \mathbf{y})$, if we specify $g_2(\mathbf{x}, \mathbf{y}) = f(\mathbf{y}) - f(\mathbf{x}) + \langle \psi(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle$ for some single valued functions $f(\cdot) : \Omega \to \mathbb{R}$ and $\psi(\cdot) : \Omega \to \mathbb{R}^m$, then (2) is reduced to the MEP studied in Kumam et al. (2010). Here we study the general case where n can be any positive integer.

Remark 2. Let $g_i(\mathbf{x}, \mathbf{y}) = f_i(\mathbf{y}) - f_i(\mathbf{x})$, $\forall i \in \mathcal{V}$ for some single valued convex function $f_i : \mathbb{R}^m \to \mathbb{R}$. In this case, variables $\mathbf{x}$ and $\mathbf{y}$ are separable. Finding $\mathbf{x}^* \in \Omega$ such that $\sum_{i=1}^n (f_i(\mathbf{y}) - f_i(\mathbf{x}^*)) \geq 0$ for any $\mathbf{y} \in \Omega$ is equivalent to solving optimization $\min_{\mathbf{x}\in\Omega} \sum_{i=1}^n f_i(\mathbf{x})$. Accordingly, MEP (2) is reduced to be a distributed optimization problem (Jakovetić et al., 2014; Nedić et al., 2010; Xi et al., 2018). Moreover, if $g_i(\mathbf{x}, \mathbf{y}) = \langle \phi_i(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle$, then MEP (2) is reduced to be a variational inequality problem.

We denote the solution set of MEP (2) by $\mathbf{X}^*$. The following assumptions are adopted throughout this paper.

Assumption 2. Set $\mathbf{X}^*$ is non-empty.

Conditions under which EPs have solutions can be found in Iusem, Kassay, and Sosa (2009).

Assumption 3. For any $i = 1, \ldots, n$, each bifunction g_i satisfies the following conditions:

(i) $\|\partial_1 g_i(\mathbf{x}, \mathbf{y})\| \le K_1$ and $\|\partial_2 g_i(\mathbf{x}, \mathbf{y})\| \le K_2$, $\forall \mathbf{x}, \mathbf{y} \in \Omega$ for some $K_1, K_2 > 0$;

(ii) $g_i(\mathbf{x}, \mathbf{y}) + g_i(\mathbf{y}, \mathbf{z}) \ge g_i(\mathbf{x}, \mathbf{z}) - c_1\|\mathbf{x} - \mathbf{y}\|^2 - c_2\|\mathbf{y} - \mathbf{z}\|^2$, $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \Omega$ for some $c_1 > 0$ and $c_2 > 0$;

In Assumption 3, condition (i) holds if Ω is compact, i.e, $\|\mathbf{x}\| \le B$, $\forall \mathbf{x} \in \Omega$, for some $B > 0$. Let $g_i(\mathbf{x}, \mathbf{y}) = \langle \phi_i(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle$, it is not difficult to verify that condition (ii) is satisfied if $\phi_i(\cdot)$ is Lipschitz continuous on Ω. Thus, condition (ii) in Assumption 3 can be viewed as a Lipschitz continuity-like condition.

Assumption 4. The global bifunction g satisfies the following conditions:

(i) $g(\mathbf{x}, \mathbf{y}) + g(\mathbf{y}, \mathbf{x}) \le 0$ for any $\mathbf{x}, \mathbf{y} \in \Omega$;

(ii) If $g(\mathbf{x}, \mathbf{y}) = g(\mathbf{y}, \mathbf{x}) = 0$ for some $\mathbf{x}, \mathbf{y} \in \Omega$, then $g(\mathbf{x}, \mathbf{z}) = g(\mathbf{y}, \mathbf{z})$ for any $\mathbf{z} \in \Omega$.

In Assumption 4, condition (i) is satisfied if $\phi_i(\cdot)$ is monotone on Ω, i.e., $\langle \phi_i(\mathbf{x}) - \phi_i(\mathbf{y}), \mathbf{y} - \mathbf{x}\rangle \le 0, \forall\, \mathbf{x}, \mathbf{y} \in \Omega$. Thus, condition (i) in Assumption 4 can be viewed as a monotonicity-like condition. Particularly, by condition (ii) in Assumption 3 and condition (i) in Assumption 4, let $\mathbf{x} = \mathbf{y} = \mathbf{z}$, it is easy to verify that $g(\mathbf{x}, \mathbf{x}) = 0$ for any $\mathbf{x} \in \Omega$. For bifunctions, condition (ii) in Assumption 4 is easy to be verified. For example, consider an EP defined as $\Omega = \{x|-1 \le x \le 1\}$ and $g(x, y) = (2|x| + |y| - 1)(|y| - |x|)$, whose solution set is $\mathbf{X}^* = \{-1/3, 1/3\}$. It is obvious that the solution to $g(x, y) = g(y, x) = 0$ is $x = \pm y, \forall y \in \Omega$. Undoubtedly, $g(x, z) = g(y, z)$ holds for any $-1 \le z \le 1$. Assumption 4 implies the cut property used in Santos and Scheimberg (2011). See the following proposition for details.

Proposition 1 (*Cut Property*). *Under Assumption 4, let $\mathbf{x}^* \in \mathbf{X}^*$, if $g(\mathbf{x}, \mathbf{x}^*) = 0$, then $\mathbf{x} \in \mathbf{X}^*$.*

Proof. By condition (i) in Assumption 4, we know that $g(\mathbf{x}, \mathbf{x}^*) + g(\mathbf{x}^*, \mathbf{x}) \le 0$. The fact $\mathbf{x}^* \in \mathbf{X}^*$ implies that $g(\mathbf{x}^*, \mathbf{x}) \ge 0$. Thus, $g(\mathbf{x}^*, \mathbf{x}) = 0$ if $g(\mathbf{x}, \mathbf{x}^*) = 0$. By condition (ii) in Assumption 4, we have $g(\mathbf{x}, \mathbf{z}) = g(\mathbf{x}^*, \mathbf{z}) \ge 0$ for any $\mathbf{z} \in \Omega$, which implies $\mathbf{x} \in \mathbf{X}^*$. $\square$

In the following subsection, we will develop a distributed algorithm to find a $x^* \in \mathbf{X}^*$.

2.3. Distributed algorithms for mixed equilibrium problems

Now consider a discrete-time multi-agent system consisting of n agents, labeled by set $\mathcal{V} = \{1, \ldots, n\}$. Each agent's dynamics is described as

$$\mathbf{x}_i(t+1) = \mathbf{u}_i(t), \qquad i \in \mathcal{V} \tag{3}$$

where $\mathbf{x}_i(t), \mathbf{u}_i(t) \in \mathbb{R}^m$, respectively, represent the state and input of agent i, and $\mathbf{x}_i(0) = \mathbf{x}_{i0} \in \Omega$ is its initial state. To solve MEP (2), the development of distributed algorithm relies on the Bregman function (Teboulle, 1997; Wang & Banerjee, 2014), which is defined as

$$D_L(\mathbf{x}, \mathbf{y}) = L(\mathbf{x}) - L(\mathbf{y}) - \langle \partial L(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \tag{4}$$

where $L(\cdot) : \Omega \to \mathbb{R}$ is a τ-strongly convex function, i.e.,

$$L(\mathbf{x}) - L(\mathbf{y}) \ge \langle \partial L(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle + \frac{\tau}{2}\|\mathbf{x} - \mathbf{y}\|^2.$$

Here we make the following assumption on $D_L(\mathbf{x}, \mathbf{y})$.

Assumption 5. $D_L(\mathbf{x}, \mathbf{y})$ is convex with respect to the second argument, i.e., $D_L(\mathbf{x}, \sum_{i=1}^n \omega_i \mathbf{y}_i) \le \sum_{i=1}^n \omega_i D_L(\mathbf{x}, \mathbf{y}_i)$, $\forall\, \mathbf{x}, \mathbf{y}_i \in \Omega$ for any positive constraints ω_i satisfying $\sum_{i=1}^n \omega_i = 1$.

It is obvious that $D_L(\mathbf{x}, \mathbf{x}) = 0$ and $D_L(\mathbf{x}, \mathbf{y}) \ge \frac{\tau}{2}\|\mathbf{x} - \mathbf{y}\|^2$ for any $\mathbf{x}, \mathbf{y} \in \Omega$. Particularly, if we set $L(\mathbf{x}) = \|\mathbf{x}\|^2$, then the Bregman function becomes the standard Euclidean distance $D_L(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$. For any $i \in \mathcal{V}$, the following distributed extragradient algorithm is proposed for MAS (3)

$$\begin{cases} \mathbf{u}_i(t) = \arg\min_{\mathbf{u}\in\Omega} \{\eta_t g_i(\mathbf{y}_i(t), \mathbf{u}) + D_L(\mathbf{u}, \mathbf{z}_i(t))\} \\ \mathbf{y}_i(t) = \arg\min_{\mathbf{y}\in\Omega} \{\eta_t g_i(\mathbf{z}_i(t), \mathbf{y}) + D_L(\mathbf{y}, \mathbf{z}_i(t))\} \\ \mathbf{z}_i(t) = \sum_{j\in\mathcal{N}_i(t)} a_{ij}(t)\mathbf{x}_j(t). \end{cases} \tag{5}$$

where $\eta_t \in \mathbb{R}$ is a positive and non-increasing step-size such that $\sum_{t=0}^{\infty} \eta_t \to \infty$ and $\sum_{t=0}^{\infty} \eta_t^2 < \infty$. The conditions for η_t are actually constraints on its decaying rate, which guarantees accurate convergence to a solution of MEP (2). This idea is inspired by the subgradient method (Boyd, Xiao, & Mutapcic, 0000; Nedić & Olshevsky, 2015). In particular, a suitable choice of η_t is $\eta_t = \frac{a_0}{t+b_0}$ for any $t \ge 0$, where a_0 and b_0 are two positive constants. By implementing algorithm (5), each agent updates its state by only using a local bifunction, its own state and states received from its neighbors. Thus, algorithm (5) is distributed.

Remark 3. The design of (5) is motivated by traditional centralized extragradient methods (Facchinei & Pang, 2003; Noor, 2003). Here Bregman function is employed to replace the squared Euclidean distance. It is worth noting that in (5), $\mathbf{y}_i(t)$ is an auxiliary variable that is determined by current states of agent i's neighbors, but not a Lagrange multiplier as in Hajinezhad, Hong, Zhao, and Wang (2016). If we let $D_L(\mathbf{x}, \mathbf{y}) = \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2$ and $g_i(\mathbf{x}, \mathbf{y}) = \langle \phi_i(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle$, then, (3) with (5) becomes a distributed double projection algorithm as follows:

$$\mathbf{x}_i(t+1) = \mathbf{P}_{\Omega}\left[\mathbf{z}_i(t) - \eta_t \phi_i\left(\mathbf{P}_{\Omega}\left[\mathbf{z}_i(t) - \eta_t \phi_i(\mathbf{z}_i(t))\right]\right)\right]$$

where $\mathbf{P}_{\Omega}[\cdot]$ represents the projection operator on Ω. This makes algorithm (5) different from the primal–dual splitting method (Hajinezhad et al., 2016) and alternating direction method of multipliers (Wang & Banerjee, 2014). Different from Facchinei and Pang (2003), Hajinezhad et al. (2016), Noor (2003) and Wang and Banerjee (2014), in this paper, we are committed to investigating the case where $g_i(\mathbf{x}, \mathbf{y})$ is nonlinear with respect to $\mathbf{x}$ and $\mathbf{y}$, and these two variables are inseparable. Moreover, the design of consensus term $\mathbf{z}_i(t)$ in (5) is inspired by consensus algorithms (Jing, Zheng, & Wang, 2017; Lu et al., 2019; Ren & Beard, 2005; Wang et al., 2004; Wang & Xiao, 2007, 2010). Since Bregman function $D_L(\cdot, \cdot)$ is strongly convex with respect to the first argument and $g_i(\cdot, \cdot)$ is convex with respect to the second argument, $\eta g_i(\mathbf{c}, \mathbf{v}) + D_L(\mathbf{v}, \mathbf{d})$ is strongly convex with respect to $\mathbf{v}$ for any $\eta \ge 0$. It implies that $\arg\min_{\mathbf{v}\in\Omega} \{\eta g_i(\mathbf{c}, \mathbf{v}) + D_L(\mathbf{v}, \mathbf{d})\}$ has a unique solution for fixed $\mathbf{c}$ and $\mathbf{d}$. Consequently, $\mathbf{u}_i$ and $\mathbf{y}_i$ in (5) exist and both uniquely exist at any t.

The goal of this paper is to achieve consensus among all agents, while ensuring that the common state is the solution to MEP (2). The definition of consensus is stated as follows.

Definition 2. MAS (3) is said to reach consensus asymptotically if $\lim_{t\to\infty} \|\mathbf{x}_i - \mathbf{x}^*\| = 0$ for any $i \in \mathcal{V}$. Moreover, $\mathbf{x}^*$ is called the consensus state.

3. Main result

In this section, we will state our main result and give its proof in detail. Let us begin with presenting the main result.

Theorem 1. *Under Assumptions 1–5, MAS (3) with (5) asymptotically reaches consensus, and the consensus state is a solution to MEP (2).*

Before giving the proof of Theorem 1, some useful lemmas will be presented. Consider the following consensus model with a perturbation:

$$\mathbf{x}_i(t+1) = \sum_{j\in\mathcal{N}_i(t)} a_{ij}(t)\mathbf{x}_j(t) + \mathbf{e}_i(t),\ \ i = 1,\dots,n \tag{6}$$

where $\mathbf{x}_i, \mathbf{e}_i(t) \in \mathbb{R}^r$. From Di Lorenzo and Scutari (2016), Nedić and Olshevsky (2015) and Zhu and Martnez (2012), we have the following lemma.

Lemma 1. *Consider model (6). Under Assumption 1, if $\|\mathbf{e}_i(t)\| \le b_t$ for some positive and non-increasing sequence $\{b_t\}_{t\in\mathbb{N}}$ satisfying $\sum_{t=0}^{\infty} b_t^2 < \infty$, then for any $i \in \mathcal{V}$,*
(i) $\lim_{t\to\infty}\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| = 0$;
(ii) $\sum_{t=0}^{\infty} b_t\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| < \infty$ where $\bar{\mathbf{x}}(t) = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i(t)$.

Next, we consider the following strongly convex optimization:

$$\min_{\mathbf{x}\in\Omega}\{f(\mathbf{x}) + D_L(\mathbf{x},\mathbf{r})\} \tag{7}$$

where $f : \Omega \to \mathbb{R}$ is convex and $\mathbf{r} \in \mathbb{R}^m$.

Lemma 2. *If $\hat{\mathbf{x}}$ is the solution to (7), then for any $\mathbf{w} \in \Omega$, there exists a subgradient $\partial f(\hat{\mathbf{x}})$ such that*

$$\langle \partial f(\hat{\mathbf{x}}), \hat{\mathbf{x}} - \mathbf{w}\rangle \le D_L(\mathbf{w},\mathbf{r}) - D_L(\mathbf{w},\hat{\mathbf{x}}) - D_L(\hat{\mathbf{x}},\mathbf{r}). \tag{8}$$

Proof. On one hand, for strongly convex optimization (7), by Karush–Kuhn–Tucker (KKT) condition, there exists a subgradient $\partial f(\hat{\mathbf{x}})$ such that

$$\langle \partial f(\hat{\mathbf{x}}) + \partial_1 D_L(\hat{\mathbf{x}},\mathbf{r}), \mathbf{w} - \hat{\mathbf{x}}\rangle \ge 0$$

which implies

$$\langle \partial f(\hat{\mathbf{x}}), \hat{\mathbf{x}} - \mathbf{w}\rangle \le \langle \partial_1 D_L(\hat{\mathbf{x}},\mathbf{r}), \mathbf{w} - \hat{\mathbf{x}}\rangle. \tag{9}$$

On the other hand, by the definition of D_L in (4), it is not difficult to verify that

$$\begin{aligned}&\langle \partial_1 D_L(\hat{\mathbf{x}},\mathbf{r}), \mathbf{w} - \hat{\mathbf{x}}\rangle\\ &= \langle \partial L(\hat{\mathbf{x}}) - \partial L(\mathbf{r}), \mathbf{w} - \hat{\mathbf{x}}\rangle\\ &= D_L(\mathbf{w},\mathbf{r}) - D_L(\mathbf{w},\hat{\mathbf{x}}) - D_L(\hat{\mathbf{x}},\mathbf{r}).\end{aligned} \tag{10}$$

Submitting (10) into (9) yields (8). □

By (3) and (5), we know $\mathbf{x}_i(t), \mathbf{y}_i(t), \mathbf{z}_i(t) \in \Omega$ for any $t \ge 0$. MAS (3) with (5) asymptotically reaches consensus. See the following lemma.

Lemma 3. *Consider MAS (3) with (5). Under Assumption 1, if $\|\partial_2 g_i(\mathbf{x},\mathbf{y})\| \le K_2$, then for any $i \in \mathcal{V}$,*
(i) $\lim_{t\to\infty}\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| = 0$;
(ii) $\sum_{t=0}^{\infty}\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\|^2 < \infty$;
(iii) $\sum_{t=0}^{\infty}\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\|^2 < \infty$;
(iv) $\sum_{t=0}^{\infty}\eta_t\|\mathbf{y}_i(t) - \bar{\mathbf{x}}(t)\| < \infty$ where $\bar{\mathbf{x}}(t) = \frac{1}{n}\sum_{t=1}^{n}\mathbf{x}_i(t)$.

Proof. (i) By the first equation in (5), we know $\mathbf{x}_i(t+1)$ is a solution to

$$\min_{\mathbf{u}\in\Omega}\{\eta_t g_i(\mathbf{y}_i(t),\mathbf{u}) + D_L(\mathbf{u},\mathbf{z}_i(t))\}.$$

Let $\hat{\mathbf{x}} = \mathbf{x}_i(t+1)$, $\mathbf{r} = \mathbf{w} = \mathbf{z}_i(t)$, using Lemma 2, we have

$$\begin{aligned}&D_L(\mathbf{z}_i(t),\mathbf{x}_i(t+1)) + D_L(\mathbf{x}_i(t+1),\mathbf{z}_i(t))\\ &\le \eta_t\langle \partial_2 g_i(\mathbf{y}_i(t),\mathbf{x}_i(t+1)), \mathbf{z}_i(t) - \mathbf{x}_i(t+1)\rangle\\ &\le K_2\eta_t\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\|.\end{aligned}$$

By the strong convexity of L, it yields

$$\begin{aligned}&D_L(\mathbf{z}_i(t),\mathbf{x}_i(t+1)) + D_L(\mathbf{x}_i(t+1),\mathbf{z}_i(t))\\ &\ge \tau\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\|^2.\end{aligned}$$

From above two inequalities, we have

$$\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\| \le K_2\eta_t/\tau. \tag{11}$$

Let $\mathbf{e}_i(t) = \mathbf{x}_i(t+1) - \mathbf{z}_i(t)$, it implies the consensus model described in (6). Note that $\|\mathbf{e}_i(t)\| \le K_2\eta_t/\tau$, using Lemma 1, we have $\lim_{t\to\infty}\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| = 0$.

(ii) By (11), we have $\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\|^2 \le (K_2\eta_t/\tau)^2$. Together with the fact $\sum_{t=0}^{\infty}\eta_t^2 < \infty$, it implies $\sum_{t=0}^{\infty}\|\mathbf{x}_i(t+1) - \mathbf{z}_i(t)\|^2 < \infty$.

(iii) Using the similar approach in (i) to the second equation in (5), it follows that $\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\| \le K_2\eta_t/\tau$. Hence, $\sum_{t=0}^{\infty}\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\|^2 < \infty$.

(iv) Note that by (iii), we have $\sum_{t=0}^{\infty}\eta_t\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\| \le \sum_{t=0}^{\infty}K_2\eta_t^2/\tau < \infty$. Furthermore, using Lemma 1, we know that

$$\sum_{t=0}^{\infty}\eta_t\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| < \infty. \tag{12}$$

Then, one has

$$\begin{aligned}&\sum_{t=0}^{\infty}\eta_t\|\mathbf{y}_i(t) - \bar{\mathbf{x}}(t)\|\\ &\le \sum_{t=0}^{\infty}\eta_t\|\mathbf{z}_i(t) - \bar{\mathbf{x}}(t)\| + \sum_{t=0}^{\infty}\eta_t\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\|\\ &\le \sum_{j=1}^{n}\sum_{t=0}^{\infty}\eta_t\|\mathbf{x}_j(t) - \bar{\mathbf{x}}(t)\| + \sum_{t=0}^{\infty}\eta_t\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\|\\ &< \infty.\end{aligned}$$ □

In Lemma 3, the statement (iii) implies $\lim_{t\to\infty}\|\mathbf{y}_i(t) - \mathbf{z}_i(t)\| = 0$. Together with the statement (i), we have $\lim_{t\to\infty}\|\mathbf{x}_i(t) - \mathbf{y}_i(t)\| = 0$ for any $i \in \mathcal{V}$. To further analyze the convergence of the proposed algorithm, we consider a Lyapunov function as follows

$$V(t) = \sum_{i=1}^{n} D_L(\mathbf{x}^*, \mathbf{x}_i(t)) \tag{13}$$

where $\mathbf{x}^* \in \mathbf{X}^*$. It is obvious that $V(t) \ge \frac{\tau}{2}\sum_{i=1}^{n}\|\mathbf{x}^* - \mathbf{x}_i(t)\|^2$. In what follows, we will analyze the convergence of $V(t)$ by studying its variation for one stage of MAS (3) with (5), which is defined as follows:

$$\Delta V(t) = V(t+1) - V(t). \tag{14}$$

Lemma 4. *Consider MAS (3) with (5). Under Assumption 5, if the communication graph $\mathcal{G}(t)$ is balanced, then,*

$$\Delta V(t) \le \eta_t \sum_{i=1}^{n} \left(g_i(\mathbf{y}_i(t), \mathbf{x}^*) - g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1))\right) - \sum_{i=1}^{n} D_L(\mathbf{x}_i(t+1), \mathbf{z}_i(t)). \tag{15}$$

Proof. By Lemma 2, let $\mathbf{r} = \mathbf{z}_i(t), \hat{\mathbf{x}} = \mathbf{x}_i(t+1), \mathbf{w} = \mathbf{x}^*$, $f(\mathbf{x}) = \eta_t g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1))$, we have

$$\begin{aligned}&\eta_t \langle \partial_2 g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1)), \mathbf{x}_i(t+1) - \mathbf{x}^* \rangle \\ &\le D_L(\mathbf{x}^*, \mathbf{z}_i(t)) - D_L(\mathbf{x}^*, \mathbf{x}_i(t+1)) \\ &\quad - D_L(\mathbf{x}_i(t+1), \mathbf{z}_i(t)).\end{aligned}$$

Together with (13) and (14), there holds

$$\begin{aligned}\Delta V(t) \le &-\eta_t \sum_{i=1}^{n} \langle \partial_2 g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1)), \mathbf{x}_i(t+1) - \mathbf{x}^* \rangle \\ &+ \sum_{i=1}^{n} (D_L(\mathbf{x}^*, \mathbf{z}_i(t)) - D_L(\mathbf{x}^*, \mathbf{x}_i(t))) \\ &- \sum_{i=1}^{n} D_L(\mathbf{x}_i(t+1), \mathbf{z}_i(t))\end{aligned} \tag{16}$$

By Assumption 5, we have

$$\begin{aligned}&\sum_{i=1}^{n} (D_L(\mathbf{x}^*, \mathbf{z}_i(t)) - D_L(\mathbf{x}^*, \mathbf{x}_i(t))) \\ &\le \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}(t)(D_L(\mathbf{x}^*, \mathbf{x}_j(t)) - D_L(\mathbf{x}^*, \mathbf{x}_i(t))) \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{n} a_{ij}(t) \right) D_L(\mathbf{x}^*, \mathbf{x}_j(t)) - \sum_{i=1}^{n} D_L(\mathbf{x}^*, \mathbf{x}_i(t)) \\ &= 0\end{aligned} \tag{17}$$

where the second equation results from the fact $\sum_{i=1}^{n} a_{ij}(t) = 1$ for any $j \in \mathcal{V}$. Moreover, by the convexity of g_i, one has

$$\begin{aligned}&\langle \partial_2 g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1)), \mathbf{x}^* - \mathbf{x}_i(t+1) \rangle \\ &\le g_i(\mathbf{y}_i(t), \mathbf{x}^*) - g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1)).\end{aligned} \tag{18}$$

Submitting (17) and (18) into (16) yields (15). □

In proving the convergence of $V(t)$, the supermartingale convergence theorem will be used, which is listed in the following lemma.

Lemma 5 (Robbins & Siegmund, 1971). *Let $\{v(t)\}_{t\in\mathbb{N}^+}$ be a non-negative scalar sequence such that*

$$v(t+1) \le (1 + a(t))v(t) - b(t) + c(t)$$

for all $t \ge 0$, if $a(t) \ge 0, b(t) \ge 0, c(t) \ge 0$ with $\sum_{t=0}^{\infty} a(t) < \infty$ and $\sum_{t=0}^{\infty} c(t) < \infty$, then the sequence $\{v(t)\}_{t\in\mathbb{N}^+}$ converges and $\sum_{t=0}^{\infty} b(t) < \infty$.

With the help of Lemmas 1–5, now we present the proof of Theorem 1.

Proof of Theorem 1. By condition (ii) in Assumption 3, we have

$$\begin{aligned}&g_i(\mathbf{z}_i(t), \mathbf{y}_i(t)) + g_i(\mathbf{y}_i(t), \mathbf{x}_i(t+1)) \\ &\ge g_i(\mathbf{z}_i(t), \mathbf{x}_i(t+1)) - c_1 \|\mathbf{z}_i(t) - \mathbf{y}_i(t)\|^2 \\ &\quad - c_2 \|\mathbf{y}_i(t) - \mathbf{x}_i(t+1)\|^2\end{aligned}$$

for any $i \in \mathcal{V}$. By (15) in Lemma 4, it follows that

$$\begin{aligned}\Delta V(t) \le &\eta_t \sum_{i=1}^{n} (g_i(\mathbf{z}_i(t), \mathbf{y}_i(t)) - g_i(\mathbf{z}_i(t), \mathbf{x}_i(t+1))) \\ &+ \eta_t \sum_{i=1}^{n} \left(c_1 \|\mathbf{z}_i(t) - \mathbf{y}_i(t)\|^2 \right. \\ &\left. + c_2 \|\mathbf{y}_i(t) - \mathbf{x}_i(t+1)\|^2 \right) \\ &+ \eta_t \sum_{i=1}^{n} g_i(\mathbf{y}_i(t), \mathbf{x}^*) \\ &- \sum_{i=1}^{n} D_L(\mathbf{x}_i(k+1), \mathbf{z}_i(t)).\end{aligned} \tag{19}$$

Applying Lemma 2 to the second equation in (5) yields

$$\begin{aligned}&D_L(\mathbf{x}_i(t+1), \mathbf{z}_i(t)) - D_L(\mathbf{x}_i(t+1), \mathbf{y}_i(t)) \\ &- D_L(\mathbf{y}_i(t), \mathbf{z}_i(t)) \\ &\ge \eta_t \langle \partial_2 g_i(\mathbf{z}_i(t), \mathbf{y}_i(t)), \mathbf{y}_i(t) - \mathbf{x}_i(t+1) \rangle \\ &\ge \eta_t (g_i(\mathbf{z}_i(t), \mathbf{y}_i(t)) - g_i(\mathbf{z}_i(t), \mathbf{x}_i(t+1)))\end{aligned} \tag{20}$$

where the second inequality holds by using the convexity of g_i with respect to the second argument. Using (19) and (20), together with the facts $D_L(\mathbf{u}, \mathbf{v}) \ge \frac{\tau}{2}\|\mathbf{u} - \mathbf{v}\|^2 \ge 0$ for any $\mathbf{u}, \mathbf{v} \in \Omega$ and $\eta_t \le \eta_0$ (η_t is non-increasing), we have

$$\begin{aligned}\Delta V(t) &\le \eta_0 \sum_{i=1}^{n} (c_1 \|\mathbf{z}_i(t) - \mathbf{y}_i(t)\|^2 + c_2 \parallel \mathbf{y}_i(t) \\ &\quad - \mathbf{x}_i(t+1) \parallel^2) + \eta_t \sum_{i=1}^{n} g_i(\mathbf{y}_i(t), \mathbf{x}^*) \\ &\le h(t) + \eta_t \sum_{i=1}^{n} g_i(\mathbf{y}_i(t), \mathbf{x}^*)\end{aligned} \tag{21}$$

where $h(t) = \eta_0 \sum_{i=1}^{n} ((c_1 + 2c_2)\|\mathbf{z}_i(t) - \mathbf{y}_i(t)\|^2 + 2c_2\|\mathbf{z}_i(t) - \mathbf{x}_i(t+1)\|^2)$ and the second inequality holds by using Young's inequality. By the boundedness of subgradient $\partial_1 g_i$, one has $g_i(\bar{\mathbf{x}}(t), \mathbf{x}^*) - g_i(\mathbf{y}_i(t), \mathbf{x}^*) \le K_1 \|\bar{\mathbf{x}}(t) - \mathbf{y}_i(t)\|$. Then,

$$\begin{aligned}&\sum_{i=1}^{n} g_i(\mathbf{y}_i(t), \mathbf{x}^*) \\ &= \sum_{i=1}^{n} g_i(\mathbf{y}_i(t), \mathbf{x}^*) - \sum_{i=1}^{n} g_i(\bar{\mathbf{x}}(t), \mathbf{x}^*) + g(\bar{\mathbf{x}}(t), \mathbf{x}^*) \\ &\le K_1 \sum_{i=1}^{n} \|\bar{\mathbf{x}}(t) - \mathbf{y}_i(t)\| + g(\bar{\mathbf{x}}(t), \mathbf{x}^*).\end{aligned} \tag{22}$$

Let $\varphi(t) = h(t) + K_1 \eta_t \sum_{i=1}^{n} \|\bar{\mathbf{x}}(t) - \mathbf{y}_i(t)\|$, combining (22) and (21) results in that

$$\Delta V(t) \le \eta_t g(\bar{\mathbf{x}}(t), \mathbf{x}^*) + \varphi(t). \tag{23}$$

By (ii)–(iv) in Lemma 3, we have $\sum_{t=0}^{\infty} \varphi(t) < \infty$. Using $g(\mathbf{x}^*, \bar{\mathbf{x}}(t)) \ge 0$ and condition (i) in Assumption 4, we have $g(\bar{\mathbf{x}}(t), \mathbf{x}^*) \le 0$. Applying Lemma 5 to (23), we know that $V(t)$ converges and $-\sum_{t=0}^{\infty} \eta_t g(\bar{\mathbf{x}}(t), \mathbf{x}^*) < \infty$. Due to the fact that $\sum_{t=0}^{\infty} \eta_t \to \infty$, we have $\liminf_{t\to\infty}(-g(\bar{\mathbf{x}}(t), \mathbf{x}^*)) = 0$. Hence, there exists a sequence $\{x(t_k)\}_{k=0,1,\ldots}$ such that $g(\lim_{k\to\infty} \bar{\mathbf{x}}(t_k), \mathbf{x}^*) = 0$. By Proposition 1, we have $\lim_{k\to\infty} \|\bar{\mathbf{x}}(t_k) - \mathbf{x}_0\| = 0$ for some $\mathbf{x}_0 \in \mathbf{X}^*$. Furthermore, by (i) in Lemma 3, there is $\lim_{t\to\infty} \|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\| = 0$. Thus, $\lim_{k\to\infty} \|\mathbf{x}_i(t_k) - \mathbf{x}_0\| = 0$. Note that $\mathbf{x}^*$ can be arbitrarily chosen from $\mathbf{X}^*$. If we let $\mathbf{x}^* = \mathbf{x}_0$, together with the convergence of $V(t)$, we have $\lim_{t\to\infty} V(t) = 0$. Recall the fact that $V(t) \ge \frac{\tau}{2} \sum_{i=1}^{n} \|\mathbf{x}_0 - \mathbf{x}_i(t)\|^2$, we finally get $\lim_{t\to\infty} \|\mathbf{x}_i(t) - \mathbf{x}_0\| = 0$ for any $i \in \mathcal{V}$. This completes the proof. □

Remark 4. Based on the proof of Theorem 1, here we analyze the convergence rate of algorithm (5) by choosing $\eta_t = (t+1)^{-(1+\alpha)/2}$ for some $0 < \alpha < 1$. To do this, it suffices to analyze the decaying rate of the deviation between $g(\mathbf{x}_i, \mathbf{x}^*)$ and zero. We use a weighted average for each $i \in \mathcal{V}$ to reflect the deviation, which is defined as: $R_i(t) = \frac{-\sum_{k=0}^{t} \eta_k g(\mathbf{x}_i(k), \mathbf{x}^*)}{\sum_{k=0}^{t} \eta_k}$. By Lemma 3 and (12), we know that there exist positive constants κ_1 and κ_2 such that $\sup \sum_{k=0}^{t} \varphi(k) = \kappa_1$ and $\sup \sum_{k=0}^{t} \eta(k)\|\mathbf{x}_i(k) - \bar{\mathbf{x}}(k)\| = \kappa_2$. Using Lipschitz continuity of $g(\cdot, \mathbf{x}^*)$, it follows that $|g(\bar{\mathbf{x}}(t), \mathbf{x}^*) - g(\mathbf{x}_i(t), \mathbf{x}^*)| \le nK_1\|\mathbf{x}_i(t) - \bar{\mathbf{x}}(t)\|$. By (23), we have $-\sum_{k=0}^{t} \eta_k g(\bar{\mathbf{x}}(t), \mathbf{x}^*) \le \sum_{k=0}^{t} \varphi(k) + V(0)$. Thus,

$$R_i(t) = \frac{\sum_{k=0}^{t} \eta_k (g(\bar{\mathbf{x}}(t), \mathbf{x}^*) - g(\mathbf{x}_i(t), \mathbf{x}^*) - g(\bar{\mathbf{x}}(t), \mathbf{x}^*))}{\sum_{k=0}^{t} \eta_k} \le \frac{V(0) + \kappa_1 + nK_1\kappa_2}{\sum_{k=0}^{t} \eta_k}.$$

By the fact that $\sum_{k=0}^{t} \eta(k) = \sum_{k=0}^{t} (k+1)^{-(1+\alpha)/2} \ge \int_0^t (k+1)^{-(1+\alpha)/2} d_k \ge \frac{2t^{(1-\alpha)/2}}{1-\alpha}$ for any $t \ge 1$, we have $R_i(t) \le \frac{(1-\alpha)(V(0)+\kappa_1+nK_1\kappa_2)}{2t^{(1-\alpha)/2}}$ for any $t \ge 1$, which implies that the convergence rate of MAS (3) with (5) is $\mathcal{O}(t^{-(1-\alpha)/2})$. Note that if α is small, then the convergence rate approximates $\mathcal{O}(t^{-1/2})$.

The balance condition of graph $\mathcal{G}(t)$ is necessary for solving MEP (2). If we remove the balance condition, algorithm (5) would be unable to solve MEP (2) even if the graph is fixed and strongly connected. However, the algorithm may still converge. Now we show convergence of the algorithm in the absence of the balance condition. For a strongly connected graph, by Perron theorem (Horn & Johnson, 2012), we know that there exists a left eigenvector $[\pi_1, \ldots, \pi_n]^T$ of weighted matrix $\mathbf{A}$ associated with eigenvalue 1 such that $\sum_{i=1}^{n} \pi_i = 1$, where $\pi_i > 0$ for all $i \in \mathcal{V}$.

Assumption 6. There exists a vector $\mathbf{x}_0 \in \Omega$ such that $\sum_{i=1}^{n} \pi_i g_i(\mathbf{x}_0, \mathbf{y}) \ge 0$ for any $\mathbf{y} \in \Omega$.

Assumption 7. $g_i(\mathbf{x}, \mathbf{y}) + g_i(\mathbf{y}, \mathbf{x}) \le 0$ for any $i \in \mathcal{V}$, $\mathbf{x}, \mathbf{y} \in \Omega$, and the equality holds if and only if $\mathbf{x} = \mathbf{y}$.

Note that Assumption 7 ensures both condition (i) and condition (ii) in Assumption 4. Under Assumptions 6 and 7, it is not difficult to verify that $\mathbf{x}_0 \in \Omega$ is the unique vector such that $\sum_{i=1}^{n} \pi_i g_i(\mathbf{x}_0, \mathbf{y}) \ge 0$, $\forall \mathbf{y} \in \Omega$.

Corollary 1. *Under Assumptions 3 and 5–7, if graph $\mathcal{G}(t)$ is fixed and strongly connected, then, MAS (3) with (5) asymptotically reaches consensus, and the consensus state $\mathbf{x}_0 \in \Omega$ satisfies $\sum_{i=1}^{n} \pi_i g_i(\mathbf{x}_0, \mathbf{y}) \ge 0$ for any $\mathbf{y} \in \Omega$.*

Proof. Consider a Lyapunov function as $V(t) = \sum_{i=1}^{n} \pi_i D_L(\mathbf{x}_0, \mathbf{x}_i(t))$. Due to the fact that $\sum_{i=1}^{n} \pi_i a_{ij} = \pi_i$ for any $j = 1, \ldots, n$, similar to (17), we have

$$\begin{aligned}
&\sum_{i=1}^{n} \pi_i (D_L(\mathbf{x}_0, \mathbf{z}_i(t)) - D_L(\mathbf{x}_0, \mathbf{x}_i(t))) \\
&\le \sum_{i=1}^{n} \pi_i \sum_{j=1}^{n} a_{ij}(t)(D_L(\mathbf{x}_0, \mathbf{x}_j(t)) - D_L(\mathbf{x}_0, \mathbf{x}_i(t))) \\
&= \sum_{j=1}^{n} \left(\sum_{i=1}^{n} \pi_i a_{ij}(t) \right) D_L(\mathbf{x}_0, \mathbf{x}_j(t)) \\
&\quad - \sum_{i=1}^{n} \pi_i D_L(\mathbf{x}_0, \mathbf{x}_i(t)) \\
&= 0.
\end{aligned} \tag{24}$$

By similar analysis as in the proofs of Lemma 4 and Theorem 1, we have

$$\begin{aligned}
\Delta V(t) &\le \eta_t \sum_{i=1}^{n} \pi_i \big(g_i(\hat{\mathbf{x}}(t), \mathbf{x}_0)\big) \\
&\le \eta_t \sum_{i=1}^{n} \pi_i \big(g_i(\hat{\mathbf{x}}(t), \mathbf{x}_0) + g_i(\mathbf{x}_0, \hat{\mathbf{x}}(t))\big) + \tilde{\varphi}(t)
\end{aligned} \tag{25}$$

where $\hat{\mathbf{x}}(t) = \sum_{i=1}^{n} \pi_i \mathbf{x}_i(t)$, and $\tilde{\varphi}(t)$ is obtained by replacing $\bar{\mathbf{x}}(t)$ in $\varphi(t)$ with $\hat{\mathbf{x}}(t)$, the second inequality holds from the fact that $\sum_{i=1}^{n} \pi_i g_i(\mathbf{x}_0, \hat{\mathbf{x}}(t)) \ge 0$. Using results in Nedić and Olshevsky (2015), Nedić et al. (2010), Ren and Beard (2005) and similar approach in Lemma 3, we can conclude that $\sum_{i=1}^{n} \tilde{\varphi}(t) < \infty$. By Assumption 7, we have $g_i(\hat{\mathbf{x}}(t), \mathbf{x}_0) + g_i(\mathbf{x}_0, \hat{\mathbf{x}}(t)) \le 0$. Applying Lemma 5 to (25), one knows that $V(t)$ converges and $-\sum_{t=0}^{\infty} \eta_t \left(\sum_{i=1}^{n} \pi_i g_i(\hat{\mathbf{x}}(t), \mathbf{x}_0) + \sum_{i=1}^{n} \pi_i g_i(\mathbf{x}_0, \hat{\mathbf{x}}(t)) \right) < \infty$. Recall that $\sum_{t=0}^{\infty} \eta_t \to \infty$, we have $\liminf_{t\to\infty} \big(-\sum_{i=1}^{n} \pi_i \big(g_i(\hat{\mathbf{x}}(t), \mathbf{x}_0) + g_i(\mathbf{x}_0, \hat{\mathbf{x}}(t))\big)\big) = 0$. This implies that there exists a sequence $\{x(t_k)\}_{k=0,1,\ldots}$ such that $\lim_{k\to\infty} \|\hat{\mathbf{x}}(t_k) - \mathbf{x}_0\| = 0$. Together with the fact that $\lim_{t\to\infty} \|\mathbf{x}_i(t) - \hat{\mathbf{x}}(t)\| = 0$, it follows that $\lim_{k\to\infty} \|\mathbf{x}_i(t_k) - \mathbf{x}_0\| = 0$. Since $V(t)$ converges, we have $\lim_{t\to\infty} V(t) = 0$. Thus, $\lim_{t\to\infty} \|\mathbf{x}_i(t) - \mathbf{x}_0\| = 0$ for any $i \in \mathcal{V}$. □

Corollary 1 implies that if the communication graph is not balanced, i.e., not every π_i is $1/n$, then MEP (2) is not solved.

4. A simulation example

In this section, we give a numerical example to illustrate Theorem 1. Given a multi-agent system with six agents, labeled by index set $\mathcal{V} = \{1, \ldots, 6\}$, the agents communicate with each other via a time-varying directed graph given in Fig. 1, where $\mathcal{G}^{(k)}$, $k = 1, \ldots, 4$ are four possible graphs and any agent is assumed to be its own neighbor. The switching order is given by $\mathcal{G}^{(1)} \to \mathcal{G}^{(2)} \to \mathcal{G}^{(3)} \to \mathcal{G}^{(4)} \to \mathcal{G}^{(1)} \to \cdots$. The weight of each edge is assumed to be $a_{ij} = \frac{1}{|\mathcal{N}_i(t)|}$, where $|\mathcal{N}_i(t)|$ is the number of agent i's neighbors. It is obvious that the union of the graphs is strongly connected with $B = 4$. We consider an electrical circuit with one-loop, which consists of six parts. The parameters in each part is detected by a specific agent. The nonlinearity of electrical devices like diodes is modeled as quadratic functions. Based on model (1), the current model is given as

$$\sum_{i=1}^{6} d_i \left(y^2 - x^{*2}\right) + (c_i x^* + b_i)(y - x^*) \ge 0, \quad \forall\, y \in \Omega$$

where $x^* \in \Omega$ represents the current in the loop. The simulation parameters are given as $\Omega = \{x | 0 \le x \le 1\}$, $d_1 = 0.3$, $d_2 = d_4 = 0.1$, $d_3 = 0$, $d_5 = 0.18$, $d_6 = 0.32$, $c_1 = 0.15$, $c_2 = 0.1$, $c_3 = 0.3$, $c_4 = 0.2$, $c_5 = 0.17$, $c_6 = 0.08$ and $b_i = 2d_i$, $i = 1, \ldots, 6$. Assume agent i has access to b_i, c_i and d_i for any $i \in \mathcal{V}$. Algorithm (5) is used for finding the solution $x^* \in \Omega$ to the MEP. By simple computation, it is easy to verify that $x^* = 0.67$ is the solution to the MEP. The initial configuration of each agent is randomly chosen from 0 to 1. By setting $\eta_t = \frac{5}{15t+2}$, the trajectories of agents' states are shown in Fig. 2, from which we see that after a period of time, the MAS almost perfectly reaches consensus, and the consensus state is close to 0.67. These observations are consistent with Theorem 1.

5. Conclusions

In this paper, a distributed extragradient algorithm has been proposed for agents to solve a class of MEPs. By implementing the

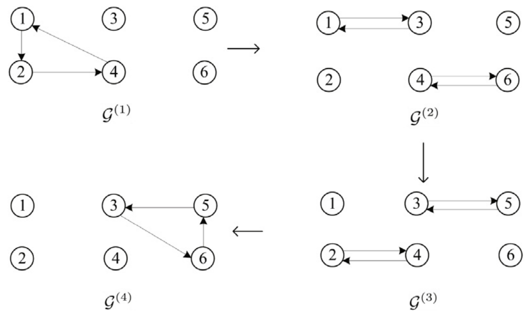

Fig. 1. The time-varying directed graph.

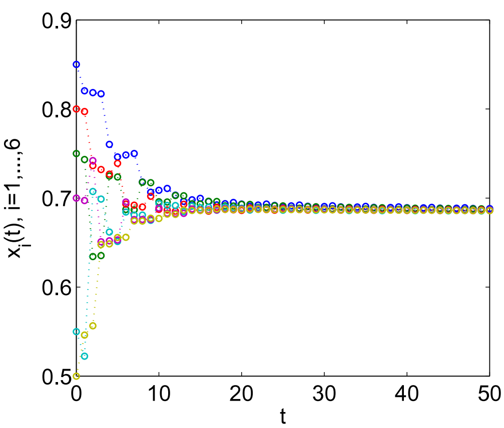

Fig. 2. Trajectories of $x_i, i = 1, \ldots, 6$ in MAS (3) with (5).

algorithm, each agent makes decisions by solving two auxiliary strongly convex optimization problems involving its own bifunction information and the local state information of its immediate neighbors. We find that if the time-varying directed graph is B-strongly connected and balanced, then the multi-agent system (MAS) reaches consensus asymptotically, and the consensus state is a solution to the MEP. The result has been proved by using graph theory, convex analysis theory and Lyapunov stability theory. A simulation example has been provided to demonstrate the effectiveness of our theoretical results. How to remove the balance condition of the directed graph and how to accelerate the convergence rate of algorithms for MEPs are two interesting and challenging problems in solving MEPs, which will be considered in our future work.

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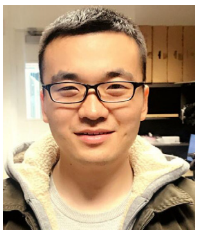

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